

Report
Of
CYBERSECURITY & ETHICAL HACKING

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By

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CANDIDATE'S DECLARATION

I, AMAN SINGH TANWAR, do hereby declare that the internship report titled "Cybersecurity & Ethical Hacking" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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I express my gratitude to my parents for their blessings.

Signature

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INTERNSHIP COMPLETION CERTIFICATE



TABLE OF CONTENTS

Chapter No.	Chapter Name	Page Nos.
--	Cover Page	--
1.	Candidate's Declaration	i
2.	Acknowledgement	ii
3.	Internship Completion Certificate	iii
4.	Project Description	1
	4.1 Introduction	1
	4.2 Organization Profile	1
	4.3 Problem Statement	2
	4.4 Project Objectives	2
	4.5 Scope of the Project	2
	4.6 Technologies and Tools Used	3
	4.7 System Architecture	3
	4.8 Methodology	4
	4.9 Expected Outcomes	6
	4.10 Certificates of Completion and Communication Proof	7
5.	Bibliography/References	8

4. Project Description

4.1 Introduction

In today's digital era, the security of information systems, networks and web applications has become one of the most critical concerns for organisations of every size. With the rapid growth of cyber threats such as data breaches, ransomware and phishing attacks, there is an increasing demand for skilled cybersecurity professionals who can identify, analyse and mitigate vulnerabilities before they can be exploited by malicious actors.

As part of the curriculum requirement of the B.Tech CSE program at IILM University, I undertook an eight-week virtual/online internship in the domain of Cybersecurity & Ethical Hacking at ApexPlanet Software Pvt. Ltd. from 21 April 2026 to 19 June 2026. The internship was designed to build strong theoretical and practical foundations in network security, web application security, penetration testing, and incident response through a structured, task-based learning approach spanning 60 days.

This report presents a detailed account of the internship, including the organisation profile, the objectives and scope of the internship, the technologies and tools used, the methodology followed across the five assigned tasks, and the key outcomes and learnings derived from the experience.

4.2 Organization Profile

ApexPlanet Software Pvt. Ltd. is a technology company that focuses on building secure, high-performance web and mobile applications by combining modern software engineering practices with cybersecurity best practices. The organisation's core philosophy centres on delivering cyber-resilient solutions through secure coding standards, encryption, and regular penetration testing to protect client businesses from evolving digital threats.

In addition to its core software development and security consulting services, ApexPlanet Software Pvt. Ltd. actively contributes to building the next generation of technology professionals by offering hands-on internship programs. These programs are structured to give students practical, industry-relevant exposure to security-first

engineering practices rather than purely theoretical instruction. More information about the organisation is available at www.apexplanet.in.

4.3 Problem Statement

Modern IT infrastructure — spanning networks, servers, and web applications — is routinely exposed to a wide range of threats including unauthorised access, injection attacks, cross-site scripting, credential theft, and denial-of-service attempts. Many organisations lack sufficient in-house expertise to proactively identify such weaknesses before they are exploited. There was, therefore, a need to develop practical, hands-on competence in identifying, exploiting (in a controlled lab environment) and remediating common security vulnerabilities across networks and web applications, and in responding effectively to security incidents when they occur.

4.4 Project Objectives

- To build strong fundamentals in cybersecurity concepts, networking, and cryptography.
- To set up and configure a professional, isolated hacking/security lab environment using virtualization tools.
- To perform reconnaissance, port and service scanning, and vulnerability assessment on target systems.
- To identify and responsibly exploit common OWASP Top 10 web application vulnerabilities in a controlled environment.
- To understand penetration testing methodology and perform exploitation and post-exploitation activities using industry-standard tools.
- To learn system hardening techniques and basic malware analysis concepts.
- To apply the acquired skills in a self-chosen capstone project and simulate an end-to-end incident response.

4.5 Scope of the Project

The scope of the internship was limited to a controlled, virtual lab environment created using virtualization software, and did not involve testing of any live, production, or third-party systems without authorisation. The work covered five broad areas over the 60-day duration: (1) foundational cybersecurity concepts and lab setup, (2) network

security and scanning, (3) web application security testing, (4) exploitation and system security using penetration testing tools, and (5) a capstone project combining these skills with a simulated incident response exercise. For the capstone, a Mini SIEM (Security Information & Event Management) solution was built using the ELK Stack (Elasticsearch, Logstash, Kibana) to centrally collect, parse, and visualise security logs from the lab environment and to detect a simulated attack. All exploitation activities were performed strictly for educational purposes on intentionally vulnerable applications such as Metasploitable2 and DVWA (Damn Vulnerable Web Application).

4.6 Technologies and Tools Used

- Virtualization: Oracle VirtualBox / VMware
- Operating Systems: Kali Linux (attacker machine), Metasploitable2 and DVWA (target/vulnerable machines)
- Network Analysis & Scanning: Nmap, Wireshark, hping3
- Vulnerability Assessment: OpenVAS / Nessus Essentials
- Web Application Testing: Burp Suite, DVWA (SQL Injection, XSS, CSRF, File Inclusion modules)
- Exploitation & Post-Exploitation: Metasploit Framework, Hydra, John the Ripper
- Cryptography Tools: OpenSSL
- Firewall & System Hardening: iptables
- SIEM (Capstone): Elasticsearch, Logstash, Kibana (ELK Stack), Filebeat
- Documentation & Version Control: GitHub, screen recording software, securityheaders.com

4.7 System Architecture

The lab environment followed a typical isolated penetration-testing architecture. A hypervisor (VirtualBox/VMware) hosted multiple virtual machines connected through a private, host-only network so that no traffic could leak to external networks. Kali Linux served as the attacker machine, pre-loaded with reconnaissance, scanning, and exploitation tools. Metasploitable2 and DVWA were configured as intentionally vulnerable target machines representing a vulnerable server and a vulnerable web

application respectively. This isolated architecture allowed reconnaissance, scanning, exploitation, and post-exploitation activities to be carried out safely without any risk to external systems.

4.8 Methodology

The internship followed a structured, task-based methodology spread across 60 days, divided into five major tasks as summarised below.

Task 1 — Foundation & Environment Setup (Days 1–12)

This phase focused on building core cybersecurity fundamentals, including the CIA Triad, common threat types (phishing, malware, DDoS, SQL Injection, brute force, ransomware) and attack vectors. The lab environment was set up by installing VirtualBox, Kali Linux as the attacker machine, and Metasploitable2/DVWA as target machines on a private host-only network. Linux fundamentals (file navigation, permissions, package management, networking commands), core networking concepts (OSI model, TCP/IP, DNS, HTTP/HTTPS, subnetting), and cryptography basics (symmetric/asymmetric encryption, hashing, SSL/TLS, hands-on OpenSSL encryption) were also studied, along with familiarisation with tools such as Wireshark, Nmap, Burp Suite, and Netcat. Deliverables included a lab setup report, a GitHub repository with Linux notes, and a short video walkthrough.

Task 2 — Network Security & Scanning (Days 13–24)

This phase covered passive and active reconnaissance techniques (Whois, Nslookup, Google Dorking, Shodan, ping sweeps, banner grabbing), followed by port and service scanning using Nmap (TCP/UDP scans, service version detection, OS detection). Vulnerability scanning was performed on Metasploitable2 using OpenVAS, and the resulting reports were analysed and categorised by severity. Packet analysis using Wireshark was carried out to capture and examine HTTP, FTP, and DNS traffic, and a simulated SYN flood attack was analysed using hping3. Basic firewall configuration using iptables was also implemented to demonstrate blocking of port scan attempts. Deliverables included an Nmap and OpenVAS scan report, a GitHub repository with scan analysis, and a demo video.

Task 3 — Web Application Security (Days 25–36)

This phase focused on identifying and understanding OWASP Top 10 vulnerabilities within the controlled DVWA lab environment, including SQL Injection, Stored and Reflected Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), and File Inclusion vulnerabilities, along with their corresponding mitigation techniques such as prepared statements, input validation, Content Security Policy, and token-based CSRF protection. Burp Suite was used for intercepting and analysing login requests and performing fuzzing with the Intruder tool, and securityheaders.com was used to evaluate HTTP security headers. Deliverables included a security testing report with screenshots, a GitHub repository documenting attack scenarios and mitigations, and a video demonstration.

Task 4 — Exploitation & System Security (Days 37–48)

This phase introduced the standard penetration testing methodology (Reconnaissance → Scanning → Exploitation → Post-Exploitation → Reporting) and its practical application using the Metasploit Framework against known vulnerabilities in Metasploitable2, including gaining system access and running post-exploitation commands. Password security was studied through brute-force SSH login attempts using Hydra and hash cracking using John the Ripper. A simulated (non-malicious) phishing awareness page was created to understand social engineering techniques and their detection. Basic malware analysis concepts (static vs dynamic analysis) and system hardening practices — applying security patches, configuring firewalls, and disabling unused services — were also studied. Deliverables included a penetration testing report, a documented GitHub repository, and a demo video.

Task 5 — Capstone Project & Incident Response (Days 49–60)

For the final phase, I chose to build a Mini SIEM (Security Information & Event Management) system using the ELK Stack (Elasticsearch, Logstash, Kibana) as my capstone project, consolidating the skills acquired over the previous four tasks into an end-to-end security monitoring and incident response exercise.

Project planning began with defining the objectives, scope, tools, and timeline, along with a network diagram showing the log sources (Kali Linux, Metasploitable2, and the DVWA web server) feeding into the central SIEM. Elasticsearch, Logstash, and Kibana were installed and configured on a dedicated VM, and Filebeat was deployed on the monitored machines to ship system, authentication, and web-server logs to the central pipeline. Logstash pipelines were built to parse and normalise the incoming log data

(e.g., SSH auth logs, Apache/DVWA access logs, and Nmap/Metasploit activity), which was then indexed in Elasticsearch.

Kibana dashboards were created to visualise key security events in near real time, including failed SSH login attempts, port-scan patterns, and suspicious HTTP requests indicative of SQL Injection or XSS attempts. Basic detection rules and alerts were configured to flag anomalous activity such as repeated authentication failures (indicating a possible brute-force attempt) and bursts of traffic consistent with a port scan or SYN flood.

To validate the Mini SIEM, a set of attacks from earlier tasks (an SSH brute-force attempt using Hydra and an Nmap port scan) were re-run against the lab targets. The resulting log activity was successfully detected and visualised on the Kibana dashboard, after which the incident response simulation was carried out: the attack was identified from the SIEM alerts and logs, the affected host was contained (blocking the offending IP via iptables), the threat was eradicated, and a post-incident report was prepared documenting the detection, containment, and remediation steps. The capstone concluded with a comprehensive final report, a GitHub repository containing the ELK configuration files, Logstash pipelines, and dashboards, and a final presentation video demonstrating the Mini SIEM detecting and responding to the simulated attack.

4.9 Expected Outcomes

- A strong practical understanding of core cybersecurity concepts, networking fundamentals, and cryptography.
- Hands-on proficiency in setting up and working within an isolated penetration-testing lab environment.
- Practical experience in reconnaissance, network scanning, and vulnerability assessment using industry-standard tools.
- The ability to identify, responsibly exploit, and mitigate common web application vulnerabilities (OWASP Top 10) in a controlled setting.
- Familiarity with the end-to-end penetration testing methodology and post-exploitation practices.

- Hands-on experience building a centralised log-monitoring solution (Mini SIEM) using the ELK Stack, including log ingestion, parsing, dashboarding, and basic alerting.
- An understanding of system hardening techniques and the basics of incident detection and response.
- Improved technical documentation, reporting, and presentation skills through structured deliverables for each task.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate issued by ApexPlanet Software Pvt. Ltd. (Certificate ID: APSPL2632834) has been attached in Chapter 3 of this report as proof of successful completion of the internship program. Copies of relevant communication and task-submission proof are maintained in the GitHub repository accompanying this report.

5. Bibliography/References

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